Tutorials and examples

SWAT+ Documentation

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These pages use the two small datasets in the `refdata` folder of the SWAT+ source repository and go through them file by file. They assume SWAT+ is installed and the Getting started pages have been read.

Pages

- Ames_sub1 walkthrough. Twelve HRUs near Ames, Iowa, simulated from 1975 to 2020, with no routing between objects. Each input group in turn, the run, and the outputs.
- Osu_1hru walkthrough. One rice paddy HRU in South Korea, simulated from 2010 to 2023, with a routing unit, an aquifer and a channel. Paddy management by decision tables, and the shipped files that 62.0.1 reads differently from their headers.

- Carbon notebook. The Jupyter notebook `jupyter_notebooks/carbon.ipynb`, which plots the soil carbon output of one HRU, and how to run it with Ames_sub1.
- Cookbook recipes. Short edits: change the simulation period, turn on an output, change the PET method, switch to Green-Ampt, add an HRU, switch on the carbon model.

Sources

- Datasets: `refdata/Ames_sub1` and `refdata/Osu_1hru` at 62.0.1.
- Notebook: `jupyter_notebooks/carbon.ipynb` at 62.0.1.

`Ames_sub1` is the reference dataset used to test SWAT+. It has twelve HRUs near Ames, Iowa, and 46 years of daily simulation (1975 to 2020), with no channels, aquifers or reservoirs. The folder is `refdata/Ames_sub1` in the source repository.

The sections below go through the input files in the order of `file.cio`, then run the model and read the outputs, as a second pass after the Quick start. Several inputs of this test dataset do not do what their names suggest. They are pointed out where they come up, because they are also the mistakes to look for in a new project.

file.cio

`file.cio` lists every other input file. Most lines in Ames_sub1 are `null`:

```
simulation        time.sim          print.prt         null              object.cnt        null
basin             codes.bsn         parameters.bsn    carbon.bsn
climate           weather-sta.cli   weather-wgn.cli   null              pcp.cli
 ↪  tmp.cli
connect           hru.con
hru               hru-data.hru
hydrology         hydrology.hyd     topography.hyd
structural        tiledrain.str     septic.str        filterstrip.str   grassedww.str
 ↪  bmpuser.str
hru_parm_db       plants.plt        fertilizer.frt    tillage.til       pesticide.pes     ...
 ↪  snow.sno
ops               harv.ops          graze.ops         irr.ops           chem_app.ops
 ↪  fire.ops   sweep.ops
lum               landuse.lum       management.sch    cntable.lum       cons_practice.lum
 ↪  ovn_table.lum
init              plant.ini         soil_plant.ini
soils             soils.sol         nutrients.sol
decision_table    lum.dtl
regions           ls_unit.ele       ...  aqu_catunit.ele
```

The only spatial objects are HRUs (`hru.con`). There is no routing: each HRU's water, sediment and nutrients leave the model, and the results are read per HRU.

Reference: file.cio.

Simulation settings

time.sim

```
day_start  yrc_start   day_end   yrc_end      step
        0       1975         0      2020         0
```

Day 0 at the start and end means 1 January of the first year and 31 December of the last. `step` 0 is a daily time step.

print.prt

```
nyskip      day_start  yrc_start  day_end   yrc_end   interval
0           0               1975         0      2020          1
...
objects                   daily       monthly        yearly         avann
hru_wb                        y             n             n             y
hru_nb                        y             n             n             y
hru_ls                        n             n             n             y
hru_pw                        y             n             n             y
hru_cb                        l             y             y             y
hru_cb_vars                   l             y             y             y
```

`nyskip` 0 means no warm-up years are left out of the outputs; for a real project set it to one to three years. Only HRU outputs are listed. Daily water balance, nutrient and plant output is on, and the carbon rows write daily output for each soil layer (`l`), which is what makes this run's output about 9 GB.

object.cnt

Twelve HRUs and no other objects.

References: time.sim, print.prt, object.cnt.

Basin settings

codes.bsn

```
pet_file  wq_file  pet  event  crack  swift_out  sed_det  rte_cha  deg_cha  wq_cha  nostress
 ↪  cn  c_fact  carbon  lapse  uhyd  sed_cha  tiledrain  wtable  soil_p  gampt  atmo_dep
 ↪  stor_max  i_fpwet  gw_flow  idc_till
null      null     2    0      0      0          0        0        0        0       0         2
 ↪  0       2       0      1     0        0          0       0       0      0         0
 ↪  0        0        3
```

- `pet` 2: potential evapotranspiration by the Hargreaves method.
- `cn` 2: read but not used in 62.0.1; the curve number always follows soil water.
- `carbon` 2: the soil carbon model is on, with its parameters from `carbon.bsn` and `carbon_lyr.bsn`.
- `gampt` 0: curve number runoff (Green and Ampt infiltration is off).

parameters.bsn and carbon.bsn

Basin-wide parameters and the carbon model coefficients. Ames_sub1 has `1.0` in column 7 of `parameters.bsn`, which switches on temperature-dependent pesticide decay.

References: codes.bsn, parameters.bsn, carbon.bsn.

Climate

`weather-sta.cli` defines one station:

```
name        wgn       pcp        tmp        slr   hmd   wnd   wnd_dir   atmo_dep
wgn_sta1    wgn_001   ames.pcp   ames.tem   sim   sim   sim   null      null
```

Precipitation comes from `ames.pcp`. Solar radiation, humidity and wind are generated (`sim`) from the monthly statistics in `weather-wgn.cli`.

Temperature does not come from a file. The station names `ames.tem`, but the temperature file list `tmp.cli` contains only `ames.tmp`, an empty file. The model cannot match the name, writes `ames.tem file not found (tgage)` to `diagnostics.out`, and generates daily temperatures from `weather-wgn.cli`. The measured record in `ames.tem` is never read. With Hargreaves PET, which depends on temperature, this affects ET as well.

hru.con gives each HRU the station name wgn_01, which is not the name in weather-sta.cli (wgn_sta1). With a single station the model uses it anyway, without a message; in a project with several stations, check that these names match.

References: weather-sta.cli, *.pcp, *.tmp, *.tem.

HRUs

hru-data.hru

Each row names the records an HRU uses:

```
id  name      topo          hydro     soil         lu_mgt            soil_plant_init  surf_stor
 ↪  snow     field
 1  hru0001   topohru0001   hyd0001   soil_01-h1   cosy_lum          soilplant1       null
 ↪  snow001  null
 2  hru0002   topohru0002   hyd0002   soil_02      mntill_corn_lum   soilplant1       null
 ↪  snow001  null
...
12  hru0012   topohru0012   hyd0012   soil_12      cosy_lum          soilplant1       null
 ↪  snow001  null
```

Eleven HRUs use cosy_lum, a corn and soybean rotation, and HRU 2 uses mntill_corn_lum, continuous corn with reduced tillage. Each HRU has a different soil.

hydrology.hyd and topography.hyd

HRU hydrology parameters (lateral flow travel time, esco, epco, cn3_swf, perco, latq_co, …) and slope, slope length and lateral flow length.

References: hru-data.hru, hydrology.hyd and topography.hyd.

Land use and management

landuse.lum links each land use to a management schedule:

Land use	Schedule in landuse.lum	Used by
cosy_lum	mgt_01	HRUs 1 and 3 to 12
mntill_corn_lum	mgt_08	HRU 2
cosy_lum2	mgt_01A	no HRU
agrr_lum, soyb_lum, oats_lum, wwht_lum	AGRR_rot, SOYB_rot, OATS_rot, WWHT_rot	no HRU; the schedules are not in management.sch

management.sch contains schedules mgt_01 to mgt_12 and mgt_01A, each a list of dated operations (tillage, planting, fertilizer, harvest). Two points about the shipped schedules at 62.0.1:

- The fertilizer name in the fert rows is one column to the right of where the model reads it, so the model finds no fertilizer and the rows apply nothing, without a message. The management page shows the expected layout.
- lum.dtl contains a decision table, irr_year_irr, meant to limit irrigation to 500 mm a year. At 62.0.1 the yearly total is not reset, so once 500 mm have been applied in total the table stops irrigation for the rest of the run.

References: landuse.lum, management.sch, *.dtl.

Databases and initial conditions

- plants.plt, fertilizer.frt, tillage.til, pesticide.pes, urban.urb, septic.sep, snow.sno: parameter databases.
- harv.ops, graze.ops, irr.ops, chem_app.ops, fire.ops, sweep.ops: parameters of each type of operation.
- plant.ini: the plant community each HRU starts with.
- soil_plant.ini: which nutrient record each HRU's soil starts with.
- soils.sol and nutrients.sol: soil layers and initial nutrient contents.

References: plants.plt, fertilizer.frt, tillage.til, *.ops, soils.sol, initial conditions.

Running the model

Check that there is about 10 GB of free space, or switch off the daily carbon output first (set the `daily` column of `hru_cb` and `hru_cb_vars` to `n` in print.prt). Then, from inside the folder:

```
path/to/swatplus-62.0.1-gnu-lin_x86_64-Rel
```

The screen shows the date being simulated:

```
  Original Simulation            1   14   1994 Yr    20 of    46 Time   12:37:36
```

The run took 4 minutes 20 seconds on a recent Linux workstation, most of it spent writing the daily carbon files.

Outputs

simulation.out

The last lines should be:

```
  Original Simulation           12   31   2020 Yr    46 of    46 Time   12:40:12

 Execution successfully completed
```

If the completion line is missing, the run stopped early; the last date shows where.

diagnostics.out

```
 DIAGNOSTICS.OUT FILE
 agrr_lum      AGRR_rot      not found in management.sch
 soyb_lum      SOYB_rot      not found in management.sch
 oats_lum      OATS_rot      not found in management.sch
 wwht_lum      WWHT_rot      not found in management.sch
 ames.tem                    file not found (tgage)
```

The first four lines are land uses whose schedules are missing; no HRU uses them, so they do not affect the results. The last line is the temperature file described under Climate.

Reference: Checker and diagnostics.

checker.out

One row per HRU with the main soil and hydrology values the model ended up with:

```
 sname        hydgrp       zmx    usle_k    sumfc     sumul   usle_p  usle_ls   esco   epco
 ↪  cn3_swf  perco  latq_co  tiledrain
soil_01-h1    B         2000.0    0.0000  361.500   616.090   1.0000   0.3638 0.9500 1.0000
 ↪  0.0000 0.5000   0.0100          0
soil_02       B         2000.0    0.0000  361.500   449.038   0.7500   0.3638 0.9500 1.0000
 ↪  0.0000 0.5000   0.0100          0
```

```
soil_03-h3    B         2000.0    0.0000  361.500   610.323   1.0000   0.3638 0.9500 1.0000
 ↪  0.0000 0.5000   0.0100          0
...
```

`usle_k` is 0 for every soil, so the HRUs produce no sediment. Checking this file for zeros and implausible values is a quick test of any new project.

hru__wb__aa.txt

The average annual water balance of each HRU. For HRU 1:

precip	snofall	surq_gen	latq	perc	et
952.6 mm	110.3 mm	34.7 mm	0.1 mm	0.0 mm	932.5 mm

Almost all precipitation leaves as ET and nothing percolates. For a real Iowa watershed, ET would be closer to two thirds of precipitation. Comparing the average annual water balance with regional estimates is the first check of any project, and soft calibration adjusts it.

Columns: Water balance outputs.

Other outputs

- `hru_nb_*`, `hru_ncycle_*`: nutrient balance and nitrogen cycle per HRU.
- `hru_ls_aa.txt`: sediment and nutrient losses per HRU.
- `hru_pw_*`: plant growth and stress per HRU.
- `crop_yld_yr.txt`, `crop_yld_aa.txt`: yields by HRU and crop.
- `mgt_out.txt`: every management operation performed, by date and HRU. Use it to check that the schedules do what was intended.
- `hru_carbvars.txt`, `hru_cflux_stat.txt`, `hru_cpool_stat.txt` and the other carbon files: see Carbon outputs.

The shipped folder also contains `basin_*` files and others that this run does not write. They are left from earlier model versions; see Quick start.

Changes to try

- Set `nyskip` to 2 in `print.prt` and compare `hru_wb_aa.txt`.
- Rename `ames.tmp` to `ames.tem` in `tmp.cli` so that the measured temperatures are used, and compare ET.
- Change the PET method in `codes.bsn` (`pet` 0 for Priestley-Taylor, 1 for Penman-Monteith); both need solar radiation, humidity and wind, which are generated here. See the cookbook.
- Follow the Carbon notebook to plot the carbon output.

Sources

- Dataset: `refdata/Ames_sub1` at 62.0.1.
- Run: the 62.0.1 `gnu-lin_x86_64` release executable, 20 October 2026. `diagnostics.out`, `checker.out`, `hru_wb_aa.txt` and the run time are from that run.
- Fertilizer column: the `fert` case of `src/mgt_sched.f90` uses `mgt%op1`, the fertilizer matched from `op_data1` (column 5); the Ames `fert` rows have `null` there and the name in `op_data2`. Details on management.sch.
- Station and file name lookups, and the irrigation decision table: see weather-sta.cli and decision tables.

`Osu_1hru` is the second reference dataset in the SWAT+ source repository. It has one HRU of 10 ha, a rice paddy in South Korea with weather from the Imsil station, simulated from 2010 to 2023. Unlike Ames_sub1, the HRU belongs to a routing unit that sends its water to an aquifer and a channel, so the routing, aquifer and channel outputs have values. The folder is `refdata/Osu_1hru` in the source repository.

The rice is managed almost entirely by decision tables: weir height, paddy irrigation, puddling, planting, fertilizer and harvest are actions in `lum.dtl`. This makes the dataset a small example of paddy management. Several shipped files are in older formats or are not read by 62.0.1, and the sections below say what the model does with each one.

file.cio

```
simulation        time.sim          print.prt         null              object.cnt        null
basin             codes.bsn         parameters.bsn    null
climate           weather-sta.cli   weather-wgn.cli   null              pcp.cli
 ↪  tmp.cli           slr.cli           null              wnd.cli           null
connect           hru.con           null              rout_unit.con     null
 ↪  aquifer.con       null              null              null        recall.con
 ↪  null              null              null              chandeg.con
channel           initial.cha       null              null              null
 ↪  nutrients.cha     channel-lte.cha   hyd-sed-lte.cha   null
```

```
reservoir          initial.res       null              null              sediment.res
 ↪  nutrients.res      weir.res              wetland.wet       hydrology.wet
routing_unit       rout_unit.def     rout_unit.ele     rout_unit.rtu     null
hru                hru-data.hru      null
...
recall             recall.rec
...
aquifer            initial.aqu       aquifer.aqu
...
water_rights       null              null              null
...
hydrology          hydrology.hyd     topography.hyd    field.fld
structural         tiledrain.str     septic.str        filterstrip.str   grassedww.str
 ↪  bmpuser.str
hru_parm_db        plants.plt        fertilizer.frt    tillage.til       pesticide.pes     null
 ↪  null               null              urban.urb         septic.sep        snow.sno
ops                harv.ops          graze.ops         irr.ops           chem_app.ops
 ↪  fire.ops           sweep.ops
lum                landuse.lum       management.sch    cntable.lum       cons_practice.lum
 ↪  ovn_table.lum
chg                cal_parms.cal     calibration.cal   null ...
init               plant.ini         soil_plant.ini    om_water.ini      null ...
soils              soils.sol         nutrients.sol     null
decision_table     lum.dtl           res_rel.dtl       null              null
regions            ls_unit.ele       ls_unit.def       null ...          aqu_catunit.ele   null
 ↪  ...
```

The `basin` line already has three slots, with `null` for the carbon file. The `water_rights` line has three `null` names where 62.0.1 reads two, and the extra `null` is ignored. Both lines are read correctly (see older file.cio files).

The `chg` line names `calibration.cal`. Its 13 rows change `cn2`, `petco`, `latq_co`, `alpha`, `epco`, `awc`, `esco`, `perco`, `cn3_swf`, `canmx`, `lat_len`, `revap_co` and `surlag` by a percentage for every object, and the model applies these changes at start-up (see hard calibration). The run below includes them.

Reference: file.cio.

Simulation settings

time.sim

```
day_start  yrc_start   day_end   yrc_end      step
        1       2010       365      2023         0
```

1 January 2010 to 31 December 2023, 14 years, with a daily time step.

print.prt

```
nyskip      day_start  yrc_start  day_end   yrc_end   interval
3           0          0          0         0         1
aa_int_cnt
0
csvout         dbout          cdfout
n              n              n
soilout        mgtout         hydcon         fdcout
n              y              n              n
objects                   daily       monthly        yearly         avann
basin_wb                     n             n              y              y
basin_nb                     n             n              n              y
```

```
...
hru_pw                          n             n             n             y
hru_cb                          n             n             n             y
hru-lte_wb                      n             n             n             y
...
pest                            n             n             n             y
```

`nyskip` 3 leaves the first three years (2010 to 2012) out of the outputs, so the average annual files cover 2013 to 2023. `basin_wb` writes yearly and average annual files, and every other row writes average annual files only.

An older SWAT+ Editor wrote this file, and 62.0.1 reads some of it differently from its headers:

- On the `csvout dbout cdfout` line, the second value is read as `use_obj_lbls`. It is `n`, so the object rows are read by position, in the fixed order listed on the print.prt page, and their names are not checked.
- On the `soilout mgtout hydcon fdcout` line, the first value is read as `crop_yld`. It is `n`, so no `crop_yld_*` files are written. `mgtout` is `y`, so `mgt_out.txt` is written.
- The fixed order has no `hru_cb` row. The `hru_cb` line is read as the `hru-lte_wb` row, and every later line is read one row early. All these lines have the same switches (`n n n y`), so in this file the shift changes nothing. The carbon rows can only be set by name, with `use_obj_lbls` = `y`.

object.cnt

```
name    ls_area  tot_area  obj  hru  lhru  rtu  mfl  aqu  cha  res  rec  exco  dlr  can  pmp
 ↪  out  lcha  aqu2d  hrd  wro
Osu          10        10    4    1     0    1    0    1    0    0    0     0    0    0    0
 ↪  0     1      0
```

Four objects: one HRU, one routing unit, one aquifer and one SWAT+ channel (`lcha`). The line ends before `hrd` and `wro`, which stay 0. `rec` is 0, so the recall object in `recall.con` is not built (see Other files).

References: time.sim, print.prt, object.cnt.

Basin settings

```
pet_file  wq_file  pet  event  crack  rtu_wq  sed_det  rte_cha  deg_cha  wq_cha  nostress  cn
 ↪  c_fact  carbon  lapse  uhyd  sed_cha  tiledrain  wtable  soil_p  gampt  atmo_dep  stor_max
 ↪  i_fpwet  gw_flow  idc_till
null      null     1    0      0      1       0        0        0        1       0         0
 ↪  0       0       0      1     0        0          0       0       0      a         0
 ↪  0        0        3
```

- `pet` 1: Penman-Monteith potential evapotranspiration, which uses solar radiation, humidity and wind.
- The sixth column has the old header `rtu_wq`, and 62.0.1 reads it as `swift_out`. Its value 1 makes every run write the SWIFT input files into a `SWIFT` folder at the end.
- `wq_cha` 1 is not used by 62.0.1.
- `carbon` 0: static soil carbon, so no carbon file is needed.
- `gampt` 0: curve number runoff.

References: codes.bsn, parameters.bsn.

Climate

`weather-sta.cli` defines one station:

```
name              wgn     pcp            tmp            slr            hmd   wnd
 ↪  wnd_dir  atmo_dep
s35610n127290e    Imsil   Imsilpcp.pcp   Imsiltmp.tmp   Imsilsol.slr   sim   Imsilwind.wnd
 ↪  null     null
```

Precipitation, temperature, solar radiation and wind come from the four Imsil files, each listed in its index file (pcp.cli, tmp.cli, slr.cli, wnd.cli). Relative humidity is generated (sim) from the monthly statistics of the Imsil record in weather-wgn.cli. The precipitation and temperature records run from 1981 to the end of 2023. The solar radiation and wind records end on day 304 of 2022, and the model generates those two variables for the rest of the run (see weather outside the observed record).

References: weather-sta.cli, *.pcp, *.tmp.

HRU, routing unit, aquifer and channel

hru.con has one HRU of 10 ha with no outflow connections (out_tot = 0):

```
id  name     gis_id  area  lat       lon        elev       hru  wst              cst  ovfl
 ↪  rule  out_tot
 1  hru0001       1    10  35.65311  127.36763  350.96629    1  s35610n127290e     0     0
 ↪  0        0
```

The HRU is the only element of routing unit rtu001 (rout_unit.def and rout_unit.ele). The routing unit collects its outflow and passes it on through rout_unit.con: surface, lateral and tile flow (tot) to channel 1, and percolation (rhg) to aquifer 1. The aquifer returns its flow to the same channel through aquifer.con. Channel 1 in chandeg.con has no outflow connection and is the basin outlet. The connection files page shows these three files.

aquifer.aqu has 21 records, aqu011 to aqu102 and aqu_deep010, all with the same parameters except the deep one. Only the first record is used: aquifer.con has one aquifer, and its aqu column points to row 1.

hru-data.hru names the records the HRU uses:

```
id  name      topo          hydro     soil        lu_mgt        soil_plant_init  surf_stor
 ↪  snow     field
 1  hru0001   topohru0001   hyd0001   PadHOEGOG   rice140_lum   soilplant1       paddy0001
 ↪  snow001  null
```

surf_stor = paddy0001 points to a record in wetland.wet whose hydrology record is paddy, which makes the HRU a paddy with ponded water and a weir. The wetland page explains the paddy records in this dataset. A misspelled name in any of the other columns would not be reported: the HRU would use the first record of that file (see name lookup).

References: *.con, hru-data.hru, aquifer.aqu, wetland.wet.

Land use and management

landuse.lum links rice140_lum to the plant community rice140_comm in plant.ini and to the schedule rice140_rot in management.sch, with curve number record rc_strow_g, conservation practice cross_slope and Manning's n record convtill_res. The plant community starts with no growing plant.

rice140_rot has one dated operation and seven decision tables:

```
rice140_rot    1    7
     plow
     weir60r
     weir0
     pad_irr_r
     puddle
     pl_hv_rice
     fert_paddy
manu      5          1      0     beef_salt   broadcast  12.9927
```

The dated operation applies 12.99 kg/ha of the manure beef_salt with broadcast on 1 May each year. The decision tables in lum.dtl do the rest:

Table	Action
plow	Tillage with fieldcultivat early in the year, once the base-zero heat units pass 0.05.
weir60r	Raises the paddy weir to 60 mm on a day between day 120 and day 130.
weir0	Lowers the weir to 0 on a day between day 270 and day 280.
pad_irr_r	From May, while the weir is up and the ponded water is shallower than 60 mm, irrigates with the ponding90 record of irr.ops.
puddle	After April, with the weir up, water ponded and less than 10 mm of rain that day, puddles the paddy with the med_eff record of puddle.ops.
pl_hv_rice	Plants rice140 with the weir up once the base-zero heat units pass 0.15, and harvests and kills it once it has reached 1.15 times its heat units.
fert_paddy	Applies 100 kg/ha of elem_n and 20 kg/ha of elem_p early in the year, and 36 kg/ha of elem_n once the rice has passed half its heat units.

Two shipped files that these tables use are not read as their authors intended:

- The plant action names the transplant record tr_rice140. That record is in transplant.plt, and 62.0.1 reads transplant records only from transplant.ops. The name is not found, no message is written, and the rice is planted from seed. mgt_out.txt still writes a TRANSPLANT line after each PLANT line. With the file renamed to transplant.ops, the model uses the record, and the planting and harvest dates change (see transplant.ops).
- puddle.ops has no title line, so the model reads its header line as the title and its first record, high_eff, as the header. The puddle table names med_eff, which is still read, so puddling works (see puddle.ops).

References: landuse.lum, management.sch, decision tables, *.ops.

Other files

- soils.sol, nutrients.sol, soil_plant.ini: soil PadHOEGOG and its initial nutrients. The soils page and the initial conditions page use Osu_1hru as their examples.
- channel-lte.cha, hyd-sed-lte.cha, initial.cha, nutrients.cha: the channel record used by chandeg.con. See SWAT+ channels.
- recall.con and recall.rec: rec is 0 in object.cnt, so no recall object takes part in the run. The model still reads recall.rec, because file.cio names it, and opens the data file RECYR. named there. That file is not in the folder, so the run leaves an empty file of that name behind. See recall.
- pesticide.pes: the half-life columns contain the old decay rates (0.007 and 0.05), which 62.0.1 reads as half-lives in days. No pesticides are simulated here, so they have no effect (see pesticide.pes).
- salt_hru.ini, salt_aqu.ini, salt_channel.ini: empty, and not used, because no salts are simulated (see salts).
- calibration.cal and cal_parms.cal: the parameter changes described under file.cio, in the format on the parameter change files page.

The shipped folder also contains output files left from runs with older revisions: mgt_out.txt, erosion.txt, fort.2222, the *.swf files at the top level and the SWIFT folder. A 62.0.1 run overwrites mgt_out.txt, erosion.txt and the files in SWIFT, and does not write the others.

Running the model

From inside the folder:

```
path/to/swatplus-62.0.1-gnu-lin_x86_64-Rel
```

With one HRU the run takes a few seconds and writes a few megabytes. The screen shows the date being simulated, and simulation.out ends with:

```
 Original Simulation           12  31  2023 Yr   14 of   14 Time  ...

 Execution successfully completed
```

Besides the output files, the run creates some empty files in the folder: `null`, `RECYR.`, `wtps_wuses.wal`, `cs_hru.ini`, `cs_aqu.ini` and `cs_channel.ini`. They come from readers that create a file when the one they look for is missing, and they do not affect the results.

Outputs

The values below are from a five-year run (2010 to 2014) of a copy of this folder, so the average annual values cover 2013 and 2014 only. A full run gives different numbers.

diagnostics.out

```
 DIAGNOSTICS.OUT FILE
```

No missing names or files are reported.

checker.out

```
sname       hydgrp      zmx   usle_k    sumfc     sumul  usle_p  usle_ls    esco    epco
 ↪  cn3_swf  perco  latq_co  tiledrain
PadHOEGOG   C         750.0   0.0000   98.114   230.336  0.7500   0.1045  0.9500  0.5000
 ↪  0.9500  0.9000   0.0100          0
```

`checker.out` is written before the changes in `calibration.cal` are applied, so `esco`, `epco`, `cn3_swf` and `perco` are the values from `hydrology.hyd`. `usle_k` is 0, so the HRU produces no USLE erosion. The sediment in the outputs comes from puddling, which sets the sediment concentration of the ponded water.

mgt_out.txt

The log of every operation and decision table action. The first season, shortened:

```
 1  2010   2  21  beef_salt   FERT
 1  2010   2  21  beef_salt   FERT
 1  2010   3  24  fldcult     TILLAGE
 1  2010   5   1  beef_salt   MANURE
 1  2010   5   4  paddy       RESET WEIR HEIGHT (m)   0.06
 1  2010   5   4  ponding     PADDY IRRIGATION
 1  2010   5   5  puddle      PUDDLE
 1  2010   5   8  rice140     PLANT
 1  2010   5   8  rice140     TRANSPLANT
 1  2010   7  21  beef_salt   FERT-WET
 1  2010  10   3  paddy       RESET WEIR HEIGHT (m)   0.00
 1  2010  10   8  rice140     HARV/KILL
```

The `FERT` lines from the `fert_paddy` table show the name `beef_salt`, the schedule's dated operation, in place of `elem_n` and `elem_p`. The amounts in the `op_var` columns (100, 20 and 36 kg/ha) are the ones applied. `FERT-WET` means the fertilizer went into the ponded water. Between planting and harvest the log also has a `PLANT_ALREADY_GROWING` line on every day the `pl_hv_rice` table finds the planting conditions met.

The 2010 rice is harvested on 8 October. In the later years of this run it does not reach 1.15 times its heat units before winter and is harvested the following spring, a few days before the next planting. Read this file to check that the decision tables do what was intended.

Water balance

`basin_wb_aa.txt` and `hru_wb_aa.txt` have the same values, because there is one HRU:

precip	irr	surq_gen	perc	et	wet_evap	wet_oflo
1321.9 mm	239.5 mm	830.8 mm	404.5 mm	333.3 mm	116.3 mm	830.1 mm

Most of the rain and irrigation leaves the paddy over the weir (`wet_oflo`), and this water is reported as surface runoff. `basin_wb_yr.txt` gives the same columns for each printed year.

Columns: Water balance outputs.

Routing, aquifer and channel

- `ru_aa.txt` and `hydout_aa.txt`: the routing unit sends about 83,000 m³ a year to the channel and about 40,000 m³ of recharge to the aquifer. The units line of these files says m³/s, but the values are volumes in m³ (see hydrographs).
- `aquifer_aa.txt`: 404.5 mm of recharge and 379.5 mm of return flow to the channel (`flo_cha`).
- `channel_sd_aa.txt`: the channel inflow is the routing unit and aquifer flow together, about 0.004 m³/s on average.

Columns: Aquifer, reservoir, plant, Channel, sediment, nutrient.

Plant growth and the paddy

- `hru_pw_aa.txt`: plant biomass, yield, stresses and weather for the HRU. With `crop_yld` set to `n` in `print.prt` there are no `crop_yld_*` files, so read the yield from this file.
- `wetland_aa.txt`: the ponded water of the paddy, with precipitation, evaporation, seepage and outflow in m³.
- `paddy_test.csv`: a daily table of the paddy (irrigation, seepage, weir height, water depth, sediment and nitrate). 62.0.1 writes it in every run.

Changes to try

- Rename `transplant.plt` to `transplant.ops` and compare the dates in `mgt_out.txt`.
- Set the first value on the `soilout` line of `print.prt` to `b` to get the `crop_yld_*` files.
- Set `pet` to 2 in `codes.bsn` for Hargreaves PET and compare `et` in `hru_wb_aa.txt`.
- Set the sixth value of `codes.bsn` to 0 to stop writing the `SWIFT` folder.
- Shorten the run in `time.sim`, or switch on daily `basin_wb` output, as shown in the cookbook.

Sources

- Dataset: `refdata/Osu_1hru` at 62.0.1.
- Run: a copy of the folder with `time.sim` shortened to 2010 to 2014 and the shipped output files removed, run with a 62.0.1 Linux build (Intel compiler) made from commit 97ca231 on 20 October 2026. That commit differs from the 62.0.1 tag only in gwflow, calibration and export coefficient code, which this dataset does not use. `diagnostics.out`, `checker.out`, `mgt_out.txt` and the output values are from that run. The transplant test used the same copy with `transplant.plt` copied to `transplant.ops`.
- `print.prt` reading: `basin_print_codes_read.f90` reads line 7 into `pco%csvout`, `pco%use_obj_labels`, `pco%cdfout` and line 9 into `pco%crop_yld`, `pco%mgtout`, `pco%hydcon`, `pco%fdcout`. In fixed order, the row after `hru_pw` is `pco%wb_sd` (`hru-lte_wb`), which receives the Osu `hru_cb` line.
- `object.cnt`: `basin_read_objs.f90` reads `bsn, sp_ob` in one list-directed read, and the short line ends the read at end of file.
- `codes.bsn`: `basin_read_cc.f90` reads the line into `bsn_cc`, whose sixth component is `swift_out`. The SWIFT files are written by `swift_output.f90`.
- `checker.out` and calibration: `checker.out` is written in `proc_hru.f90`, and `proc_cal` runs later in `main.f90.in`.
- Transplant lookup: `dtbl_lum_read.f90` matches the `plant` action's file pointer against `transpl%name`, which `plant_transplant_read.f90` fills from `transplant.ops` only. In the `plant` case of `actions.f90`, `mgt_transplant` is called only when `act_app` > 0, but the `TRANSPLANT` line is written whenever `pco%mgtout` = `y`.
- `fertilize` action log: the `fertilize` case of `actions.f90` writes `mgt%op_char` in the name column of `mgt_out.txt`, where `mgt` is the HRU's current schedule operation (`sched%mgt_ops(hru%cur_op)`, set in `mgt_operatn`).

- Recall: recall_read.f90 reads recall.rec whenever the file exists, independent of the rec count, and opens each record's data file without checking that it exists, which creates RECYR..
- Puddle file: mgt_read_puddle.f90 skips a title and a header line before the records.

The SWAT+ source repository has one analysis notebook, jupyter_notebooks/carbon.ipynb. It reads the carbon output of one HRU and draws time series of soil carbon, the carbon pools by soil layer, plant carbon and the carbon fluxes, with the HRU's management operations marked on each graph. It uses the legacy CSU carbon files, which have no interval suffix in their names (see legacy CSU carbon outputs). The newer carbon files, such as hru_cbn_lyr_day.txt, are not read by the notebook.

Files read

The notebook reads these files from one folder:

File	Use
hru-data.hru, landuse.lum	Soil, land use and schedule names of the chosen HRU.
mgt_out.txt	Management operations of the HRU, drawn as markers.
hru_cbn_lyr.txt	Total soil carbon by layer.
hru_seq_lyr.txt	Sequestered soil carbon, total and in the top 300 mm.
hru_plc_stat.txt	Plant carbon: above ground, leaf, stem, seed, root and surface residue.
hru_cpool_stat.txt	Residue, metabolic, structural, lignin, microbial, slow humus and passive humus carbon, root mass and soil water, by layer.
hru_org_trans_vars.txt	Potential transformation rates (bmctp, hsctp, hsntp, hpctp, lslctp, lslnctp) by layer.
hru_cflux_stat.txt	Carbon flows between pools and CO2 losses by layer.

The input files and the output files must be in the same folder, which is the case unless file.cio sets an output folder.

Switches

The legacy files are written only with these settings:

- carbon = 2 in codes.bsn, with carbon.bsn and carbon_lyr.bsn present (see carbon model). hru_plc_stat.txt, hru_cpool_stat.txt, hru_cflux_stat.txt and hru_org_trans_vars.txt are written only with carbon = 2.
- use_obj_lbls = y in print.prt, because the carbon rows are recognised only by name.
- A hru_cb row with y or l in the daily column, for the first five files in the table. l adds layer properties at the start and end of the run.
- A hru_cb_vars row with y or l in the daily column, for hru_org_trans_vars.txt.
- mgtout = y in print.prt, for mgt_out.txt.

The notebook uses the shortest interval in each file, and it is designed for daily output.

Ames_sub1 setup

The Ames_sub1 reference dataset at 62.0.1 already has all of these settings:

```
csvout        use_obj_lbls  cdfout
n             y             n
crop_yld      mgtout        hydcon        fdcout
b             y             n             n
objects                  daily       monthly        yearly         avann
...
hru_cb                     l             y             y             y
hru_cb_vars                l             y             y             y
```

The full run of 46 years writes about 9 GB, most of it daily carbon output. For the notebook, shorten the run first: with `time.sim` set to 1975 to 1976, the run takes a few seconds and the output folder is about 380 MB.

Running the notebook

1. Copy `refdata/Ames_sub1` to a new folder, shorten `time.sim`, and run SWAT+ in the copy (see Running the model).
2. Install the Python packages. The notebook uses `pandas` and `plotly`, and plotly needs `nbformat` to show figures in Jupyter:

   ```
   pip install jupyterlab pandas plotly nbformat
   ```

3. Open `carbon.ipynb` in JupyterLab or VS Code.
4. In the first code cell, under `# hru to run against.`, set:
   - `hru_id`: the HRU number to plot (1 by default).
   - `wdir` and `sdir`: the notebook reads from the folder `f"{wdir}/{sdir}"`. The values in the shipped notebook are the author's folders and must be changed, for example `wdir = "/home/me/runs"` and `sdir = "Ames_sub1"`.
   - `output_html_file`: `True` also saves every figure as an HTML file.
5. Run all cells.

The cell that reads a file does nothing when the file is missing, and the graph cells after it then work with the table of the previous file, or stop with an error if there is none. Turn on every switch listed above before running the notebook.

Graphs

Each section of the notebook has a cell that reads one file and one or more cells that draw Plotly figures from it. With daily output for HRU 1 of Ames_sub1, the notebook draws, in order:

1. Total soil carbon by layer (`hru_cbn_lyr.txt`), in Mg/ha per mm of layer thickness.
2. Sequestered carbon in the top 300 mm (`hru_seq_lyr.txt`), in Mg/ha.
3. Plant carbon components (`hru_plc_stat.txt`), in kg/ha.
4. Residue and soil carbon pools in the 10 mm top layer, and in the layer at a depth set in the cell (100 mm by default) (`hru_cpool_stat.txt`).
5. One graph by layer for each pool in `hru_cpool_stat.txt`: residue, metabolic, structural, lignin, microbial, slow humus, passive humus, live root mass and soil water content.
6. One graph by layer for each potential transformation rate in `hru_org_trans_vars.txt`.
7. One graph by layer for each carbon flow between pools (`cfmets1` to `cfs3s1`) and each CO2 loss (`co2fmet` to `co2fs3`) in `hru_cflux_stat.txt`.

Several graph cells have a `normalize` setting that divides each layer's value by the layer thickness, and a `show_mgt` setting that turns the management markers on or off. The markers are coloured by operation type: planting, harvest, tillage, fertilizer, manure, irrigation, pesticide and grazing.

Related pages

- Carbon outputs: the newer carbon files and the legacy files with their columns.
- carbon.bsn: carbon model parameters and initial pools.
- Soil carbon theory: what the pools and flows represent.
- Cookbook: carbon model: switching the carbon model on in another project.

Sources

- Notebook: `jupyter_notebooks/carbon.ipynb` at 62.0.1. The file names, the folder variables `wdir` and `sdir`, the `read_data` and `read_mgt_out` functions and the graph cells are in its code cells.

- Test: a copy of `refdata/Ames_sub1` with `time.sim` set to 1975 to 1976, run with a 62.0.1 Linux build (Intel compiler) made from commit 97ca231, which differs from the tag only in gwflow, calibration and export coefficient code. The run took about 10 seconds and wrote 379 MB. All code cells of the notebook were then run outside Jupyter with pandas 3.0.1 and plotly 6.6.0, with `wdir` pointed at the copy; every cell completed and 33 figures were drawn, with 731 daily values (1975 and 1976) in each data trace.
- Legacy file switches: `command.f90` calls `soil_nutcarb_write_legacy.f90` for `pco%cb_hru` and `soil_carbvar_write_legacy.f90` for `pco%cb_vars_hru` (only with `bsn_cc%cswat` = 2). The files are opened in `carbon_legacy_module.f90`. The `freq` column takes the values `day`, `mon` and `year`, which the notebook's reading cells filter on.
- Missing files: each reading cell tests `file_path.is_file()` and otherwise only sets `df_plot` to an empty table; the graph cells test `df.empty`, where `df` is the table from the last reading cell that found its file.

Six common edits to a SWAT+ project that already runs. Each recipe names the file and the values to change, and what to check after the run. The input reference pages linked from each recipe give the full rules.

Simulation period

Edit `time.sim`:

```
day_start  yrc_start   day_end   yrc_end      step
        0       1975         0      2020         0
```

- `yrc_start` and `yrc_end` are the first and last calendar years.
- `day_start` is the day of the year the run starts; 0 is read as 1, 1 January.
- `day_end` is the day of the year the run ends in the last year; 0 means 31 December.

To leave warm-up years out of the outputs, keep `time.sim` and set `nyskip` on line 3 of `print.prt` to the number of years to skip. The skipped years are still simulated.

Days outside the measured weather records are generated from `weather-wgn.cli` without a message, so a period longer than the records still runs. Compare the period with the first and last dates in the `.pcp` and `.tmp` files before reading the results.

References: time.sim, print.prt, weather outside the observed record.

Daily or monthly output

Each object row of `print.prt` has four switches, for daily, monthly, yearly and average annual files. Set the column to `y` to write that file, for example daily basin water balance:

```
objects                 daily       monthly        yearly         avann
basin_wb                    y             n             y             y
```

How the rows are read depends on the second value on line 7 (`use_obj_lbls`, labelled `dbout` in files from older Editors):

- `y`: rows are matched by name. A row can be added, removed or moved, and the carbon and gwflow rows can be used.
- `n`: rows are read by position in a fixed order. Do not add, remove or reorder rows: every later row would apply to the wrong output, without a message.

Daily output of many objects over a long period can take several gigabytes, so check the free disk space before a long run. After the run, `files_out.out` lists every output file that was opened.

References: print.prt, Output naming convention.

PET method

Edit `pet`, the third value in `codes.bsn`:

`pet`	Method	Weather it uses
0	Priestley-Taylor	temperature and solar radiation
1	Penman-Monteith	temperature, solar radiation, relative humidity and wind
2	Hargreaves	temperature
3	weather station PET	in 62.0.1 this gives Hargreaves PET (see below)

The PET method does not change which weather files are read. A variable that the station takes from the weather generator (`sim` in `weather-sta.cli`) is generated in every case, so Penman-Monteith with generated humidity and wind uses generated values. With `pet` = 3, 62.0.1 replaces the measured station PET with Hargreaves PET before the HRUs use it, so this code gives Hargreaves PET times `pet_co`.

After the run, compare `pet` and `et` in `hru_wb_aa.txt` or `basin_wb_aa.txt` with the earlier run.

References: PET method, weather-sta.cli.

Green-Ampt infiltration

Three files change:

1. `codes.bsn`: set `gampt`, the 21st value, to 1.
2. `time.sim`: set `step` to the number of time steps per day, for example 24 for hourly steps.
3. The precipitation files: give sub-daily records with the same number of values per day as `step`. With `step` above 1, 62.0.1 does not split a measured daily record over the time steps, so a daily `.pcp` file gives no rain in any time step.

Plant growth, soil water and nutrient cycling still run once a day.

After the run, compare `surq_gen` and `perc` in `basin_wb_aa.txt` with the curve number run.

References: runoff method, time step, *.pcp.

HRU addition

An HRU needs a row in several files. Copy the rows of an existing HRU and change them:

1. `hru.con`: add a row with the next `id`. Rows must be numbered 1, 2, 3 and so on in file order. Set `area`, `lat`, `elev`, the `hru` column (the row number of the new HRU in `hru-data.hru`) and `wst` (a station name in `weather-sta.cli`).
2. `hru-data.hru`: add a row whose `topo`, `hydro`, `soil`, `lu_mgt`, `soil_plant_init`, `snow` and `field` names exist in the corresponding files. A misspelled name is not reported: the HRU uses the first record of that file.
3. The HRU's outflow: either give it outflow connections in `hru.con` (`out_tot` and the groups after it), or add it as an element of a routing unit in `rout_unit.ele` and the element list in `rout_unit.def`, as in Osu_1hru. With neither, its water leaves the model.
4. `ls_unit.ele` and `ls_unit.def`, if the project has them: add the HRU as an element. The basin output files (`basin_wb` and the others) are added up from these elements with their `bsn_frac` values as given, so the `bsn_frac` values should add up to 1 after the change.
5. `object.cnt`: add 1 to `hru` and to `obj`.

After the run, `checker.out` has one row per HRU, in HRU order, with the soil and hydrology values the new HRU ended up with. Check `diagnostics.out` for names that were not found, and remember that the `topo`, `hydro`, `soil`, `lu_mgt`, `soil_plant_init` and `snow` names are not checked there.

References: *.con, hru-data.hru, object.cnt, print.prt.

Carbon model

The dynamic soil carbon model needs three changes:

1. codes.bsn: set carbon, the 14th value, to 2. The value 1 is reserved for C-FARM, which is not implemented, and runs the model with no organic matter cycling at all. Datasets from before revision 62 used 1 for the dynamic model and must be changed.
2. file.cio: name the carbon file as the third name on the basin line, for example basin codes.bsn parameters.bsn carbon.bsn.
3. Provide carbon.bsn and carbon_lyr.bsn in the project folder. refdata/Ames_sub1 has both, and they can be copied as a start. SWAT+ stops with an error if either is missing. The old carb_coefs.cbn is no longer read.

For carbon output, print.prt must use use_obj_lbls = y, and the carbon rows are added by name. The newer rows (hru_cb_gl, hru_cb_lyr, hru_cb_cpool and others) each switch one family of files, and the legacy rows hru_cb and hru_cb_vars switch the files the carbon notebook reads. Daily per-layer carbon output is large: in Ames_sub1 it makes up most of the 9 GB of a full run, so start with a short period.

After the run, files_out.out lists the carbon files that were opened, and Carbon outputs describes their columns.

References: carbon model, carbon.bsn, older file.cio files, carb_coefs.cbn.

Sources

- Simulation period: time_read.f90 sets time%day_start to 1 when it is 0 or less.
- Output rows: basin_print_codes_read.f90 reads the second value of line 7 into pco%use_obj_labels and the object rows by position or by name accordingly.
- codes.bsn: basin_read_cc.f90. Column positions: pet 3 (bsn_cc%pet), carbon 14 (bsn_cc%cswat), gampt 21 (bsn_cc%gampt). PET with pet = 3 and sub-daily precipitation: see the Sources sections of codes.bsn and time.sim.
- HRUs: connection rows are read by hyd_read_connect.f90, HRU rows and their name lookups by hru_read.f90, counts by basin_read_objs.f90, and checker.out is written by proc_hru.f90. Basin output weighting by bsn_frac: see nutrient balance and losses.
- Carbon: the basin line of file.cio is read in readcio_read.f90, and carbon.bsn and carbon_lyr.bsn by carbon_bsn_read.f90.